Anushka Ghei

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EDUCATION

MS in Computer Science and Engineering, University at Buffalo, SUNY — Buffalo, NY | Expected Feb 2027
Specialisation: Artificial Intelligence and Machine Learning. GPA: 3.7/4
Coursework: Deep Learning, Machine Learning, Operating Systems, Data Intensive Computing, Data Models and Query Languages, Computer Vision

B.Tech in Computer Science and Engineering, PES University — Bangalore, India | Sep 2021 - May 2025
Specialisation: Machine Intelligence and Data Science. GPA: 8.33/10

WORK EXPERIENCE

Software Engineering Intern, Offerwell — Buffalo, NY | Jun 2026 - Present

- Built core features of a real-estate transaction platform in Next.js, TypeScript, and PostgreSQL: paginated Kanban boards and no-code intake forms that let brokerages customize workflows without engineering help.
- Built a Gemini LLM pipeline that extracts structured data from purchase contracts to prefill transactions, with per-field confidence and source citations; fixed production failures on large forms via parallel schema batching.
- Fixed multi-tenant access-control flaws that exposed other agents' deals and client PII, enforcing server-side authorization across six API surfaces including mobile endpoints, webhooks, and cron jobs.
- Replaced in-memory JavaScript aggregation with SQL GROUP BY, removing a 5,000-row cap that under-reported large brokerages' sales; added idempotency keys so retried writes never create duplicates.

Senior Research Aide, Weiss Labs, University at Buffalo — Buffalo, NY | Jan 2026 - Present

- Built the analysis pipeline for an NIA-funded sleep and cognition trial, turning multi-stream Fitbit telemetry into 12-week participant trajectories.
- Engineered circadian and sleep-architecture biomarkers from raw actigraphy into a composite Alzheimer's-risk index using K-means clustering.
- Modeled sleep and cognition associations across 1,255 MIDUS participants with mixed-effects regression and Random Forest, interpreted via SHAP.

Software Engineering Intern, Siemens Healthineers — Bangalore, India | Jan 2025 - Jul 2025

- Designed and deployed a scalable Linux data pipeline on RKE2 and Kubernetes clusters backed by PostgreSQL, automating multi-format output generation with Trivy and CI/CD tooling on an Agile Scrum team.
- Automated metadata retrieval and enrichment against GitHub and GitLab package registry APIs, increasing coverage by 45% and accuracy by 40%.

Software Development Intern, Opkey — Bangalore, India | Aug 2024 - Oct 2024

- Fine-tuned LLMs as ReAct agents and built agentic workflows chaining model inference with dynamic tool execution, improving method-invoking accuracy by 40%.
- Implemented an automated API testing framework in Postman and cURL validating responses and token authentication for internal ML tools, wired into CI/CD.

AI and Data Intern, Deloitte — Bangalore, India | Jun 2024 - Aug 2024

- Automated a geospatial data pipeline and benchmarked clustering for a recommender, improving SKU targeting by 65%.

Data and Analytics Intern, PwC — Bangalore, India | Jun 2023 - Jul 2023

- Built Power BI dashboards tracking ad KPIs across campaigns and benchmarked Indian e-commerce platforms.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, C#, Scala, SQL, HTML/CSS
AI/ML and LLMs: PyTorch, HuggingFace Transformers, scikit-learn, RAG, agentic workflows, prompt engineering, OpenAI/Anthropic/Gemini APIs, FAISS, TensorFlow, SHAP
Backend and Data: Node.js, REST APIs, .NET, PostgreSQL, MongoDB, MySQL, Apache Spark, Hadoop, Power BI
Frontend: React, React Native, Next.js
DevOps and Infra: Docker, Kubernetes, RKE2, CI/CD, AWS, Linux, Git, Postman, Trivy, Claude Code, Cursor, GitHub Copilot

PROJECTS

Detecting Hallucinations in LLMs | *Python, PyTorch, FAISS, RAG*
Built a 2-agent LLM system that flags self-contradictions and factual inaccuracies, cutting the hallucination rate by 25% through Step-back Prompting and Retrieval-Augmented Generation (RAG).

Yet Another Distributed File System | *Python, Apache Spark, Hadoop, Sockets, Multi-threading*
Built a master-slave distributed file system with replication factor 3 for high availability and fault tolerance across a network.

PUBLICATIONS

Five 2024 papers on NLP question generation, multimodal sentiment analysis, underwater object detection, and crowd simulation.